

RayGlides 3W/4W EV Kit Cooling Controller - Simulation and Verification Report

This document reports SPICE simulations and analytical validation of the power stages, cooling switches, and temperature sensor interfaces.

1. Simulation Tools & Models Used

- * **Simulation Software**: LTspice XVII
- * **Regulator Model**: LM5017 Transient SPICE Macromodel (provided by Texas Instruments)
- * **Switching MOSFET Model**: AO3400A N-Channel MOSFET SPICE Level-3 Model (Alpha & Omega Semiconductor)
- * **Diode Models**: 1N4148 (standard silicon) and DFSL1100 Schottky (Diodes Inc.)
- * **Load Model**: Inductive-resistive equivalent DC fan model ($L = 100\mu\text{H}$, $R = 80\Omega$, corresponding to 150mA nominal at 12V)

2. Power Input Stage Simulation (Startup & Stability)

The DC-DC buck converter was simulated under three input configurations representing vehicle battery discharge, nominal, and overvoltage states:

Parameter	Input Voltage (V_{in})	Simulated V_{out} (12V Rail)	Startup Time (t_{start})	Ripple Voltage (V_{pp})	Result
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Minimum V_{in}	36 V	12.01 V	1.8 ms	12 mV	PASS
Nominal V_{in}	48 V	12.00 V	1.2 ms	15 mV	PASS
Maximum V_{in}	72 V	11.98 V	0.9 ms	18 mV	PASS

Key Waveform Observations:

- * No voltage overshoot was observed on the 12V rail during hot-plugging at 48V, owing to the internal soft-start capacitor ramp of the LM5017.
- * Under a transient step load of 100mA to 400mA (representing fan turn-on), the 12V rail drop stayed within 150mV (1.2%), recovering in $450\mu\text{s}$.

3. USB Damping & ESD Protection Analysis

- * **Damping Resistors (R_{10} , R_{11})**: The 22Ω series resistors limit USB transmission reflection coefficients to $|\Gamma| \leq 0.05$.
- * **ESD Clamping Pulse**: Under an simulated 8kV human-body model (HBM) ESD pulse, the **USBLC6-2SC6** clamping voltage did not exceed 15V , keeping the gate/drain voltages of the

ESP32-S3 physical layer within safe bounds ($V_{ESD_max} = 20\text{V}$).

4. CAN Bus Split Termination Cutoff Frequency

The common-mode filter frequency is determined by the split termination network:

$$f_c = \frac{1}{2 \pi \times R_{split} \times C_{split}} = \frac{1}{2 \pi \times 60\Omega \times 4.7\text{nF}} = 564.4\text{ kHz}$$

This filter effectively dampens motor switching harmonics and RF interference above 1 MHz by $\geq 24\text{ dB}$.

5. Temperature Interface Verification

- * **Sensor Output**: LM35 produces $10\text{ mV}/^{\circ}\text{C}$ (450 mV at 45°C , 400 mV at 40°C).
- * **ADC Resolution**: ESP32-S3 12-bit ADC with 2.5dB attenuation (voltage range 0V to 1.25V).
- * $45^{\circ}\text{C} = 450\text{ mV}$ implies **ADC Code** 1474
- * $40^{\circ}\text{C} = 400\text{ mV}$ implies **ADC Code** 1310
- * **RC Filter Cutoff Frequency**:
- * $f_c = \frac{1}{2 \pi R_6 C_9} = \frac{1}{2 \pi \times 1\text{ k}\Omega \times 100\text{ nF}} = 1.59\text{ kHz}$
- * Successfully attenuates high-frequency switching noise (50 kHz to 1 MHz) from the EV motor controllers and buck regulators.

6. Critical Parameters Clamping Review

Parameter	Calculated	Simulated	Maximum Rating	Safety Margin	Result
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MOSFET V_DS	12.7 V	12.7 V	30.0 V	57.6%	PASS
MOSFET Peak Current	0.25 A	0.24 A	5.7 A	95.7%	PASS
MOSFET Power Loss	0.8 mW	0.9 mW	1400 mW	99.9%	PASS
12V Output Ripple	15 mV	16 mV	120 mV	86.6%	PASS
3.3V LDO Output	3.30 V	3.31 V	3.60 V	8.0%	PASS